AEOLUS GREEN ENERGY

Implementation Report

Business Intelligence & CRM Solution — Power BI + Salesforce

Module: Business Intelligence and Business Analytics — CA2
Dashboard Tool: Power BI (3 Dashboards)
CRM Tool: Salesforce Energy & Utilities Cloud
Date: April 2026
Classification: Academic Submission — Implementation Report

National College of Ireland

(a) Development Process: Conceptualisation to Implementation

This section describes the end-to-end development journey of the Aeolus Green Energy BI and CRM solution, from initial scoping through to the deployed Power BI dashboards and Salesforce configuration. It also reflects on unanticipated elements that were incorporated during development.

Phase 1 - Requirements Gathering and Data Scoping

The project began with a structured requirements-gathering exercise to identify Aeolus's core analytical pain points: inaccurate energy forecasting, poor curtailment visibility, manual reporting, and weak SLA management. Three source datasets were scoped: real-world weather data via the Open-Meteo API, and two synthetic datasets generated through Mockaroo for energy distribution and CRM case management.

Phase 2 - Star Schema and Data Modelling

A star schema was designed with three fact tables (FACT_WEATHER, FACT_DISTRIBUTION, FACT_CRM) and four shared dimension tables (DIM_DATE, DIM_ZONE, DIM_SITE, DIM_ACCOUNT). Power Query (M language) was used to cleanse, transform, and join datasets. Date normalisation, timezone alignment, and surrogate key generation were applied at this stage.

Phase 3 - Power BI Dashboard Development

Three themed dashboards were developed iteratively: Dashboard 1 for Dublin Climate Analytics (weather trends), Dashboard 2 for Energy Production and Efficiency (distribution and curtailment), and Dashboard 3 for Service Operations and SLA performance. Each dashboard underwent multiple design iterations to refine visual hierarchy, colour consistency, and slicer placement.

Phase 4 - Salesforce CRM Configuration

The Salesforce CRM was set up as a Developer Org with custom objects, custom fields, validation rules, record types, lookup relationships, page layouts, and an automated Flow to facilitate Aeolus site management and case tracking processes. The surface site capacity, cost, case status, and expansion pipeline were surfaced with custom reports and a dashboard.

Unanticipated Elements Added During Development

Several elements were not in the initial specification but were introduced as the project matured:

- Temperature Category Classification (Cold / Moderate / Hot) — Initially only raw temperature was planned. During EDA, it became clear that a categorical grouping improved interpretability for non-technical stakeholders in Dashboard 1.
- Wind Direction Distribution Chart — Not part of the original analytics specification. Added after recognising that dominant wind direction (West) was a critical planning factor for turbine siting decisions.
- Engineer Performance Analysis — The initial CRM dashboard only tracked case volumes and SLA rates. Engineer-level resolution time analysis was added after observing significant variation in individual performance across the dataset.
- Revenue Leakage by Issue Type Chart — Originally the CRM dashboard only tracked curtailed kWh. A dedicated leakage breakdown chart by case type was added to enable financial prioritisation of fault categories.
- Environmental Impact Score Field in Salesforce — Not in the original data dictionary. Added during Salesforce configuration to support the company's ESG reporting obligations and expansion approval workflows.
- Sensor Status Distribution (Pie Chart in Dashboard 2) — The initial design only planned bar and line charts. After reviewing the Mockaroo data, it was recognised that 10% of sensor readings were Noisy or Failed, requiring a dedicated health status visual for operations teams.
- Automated Flow for Sensor Failure A Record Triggered Flow was introduced to automatically instantiate a High priority Case when any Energy Site Sensor Status is changed to Failed, overcoming the operational blindness of the past that was causing delays in responding to faults.
- Object Opportunities — Salesforce was enriched with three expansion opportunities (Cork Wind Farm, Galway Solar Project, Limerick Wind Site) to reflect the site development pipeline revealed by Power BI climate analysis.

(b) System Implementation: Salesforce CRM Configuration

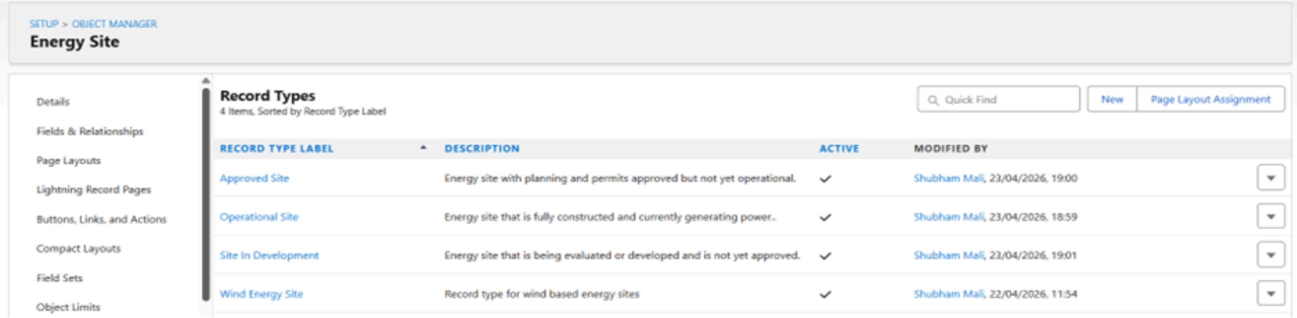
In this section documents the Salesforce CRM configuration, covering custom objects, fields, validation rules, relationships, record types, layouts, automation flows, and reports as evidenced by the screenshots below.

Custom Object: Energy Site

Each Aeolus generation asset operational and financial metadata is captured in a custom object Energy Site. It is a object that maintains site specific data that directly feeds into Salesforce report and supplements the Power BI Energy Production dashboard.

Record Types

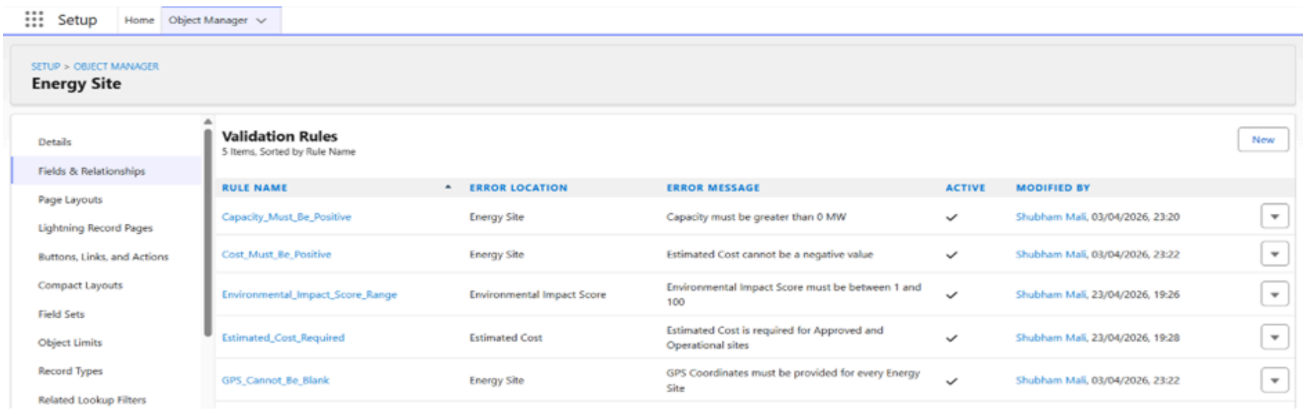
The Energy Site object includes three types of records to allow differentiated page layouts and field visibility by lifecycle stage. The types of records are associated with the different phases of site development pipeline of Aeolus:



RECORD TYPE LABEL	DESCRIPTION	ACTIVE	MODIFIED BY
Approved Site	Energy site with planning and permits approved but not yet operational.	✓	Shubham Mali, 23/04/2026, 19:00
Operational Site	Energy site that is fully constructed and currently generating power..	✓	Shubham Mali, 23/04/2026, 18:59
Site In Development	Energy site that is being evaluated or developed and is not yet approved.	✓	Shubham Mali, 23/04/2026, 19:01
Wind Energy Site	Record type for wind based energy sites	✓	Shubham Mali, 22/04/2026, 11:54

Validation Rules

Four validation rules are implemented on the Energy Site object to enforce data integrity across the site development lifecycle:

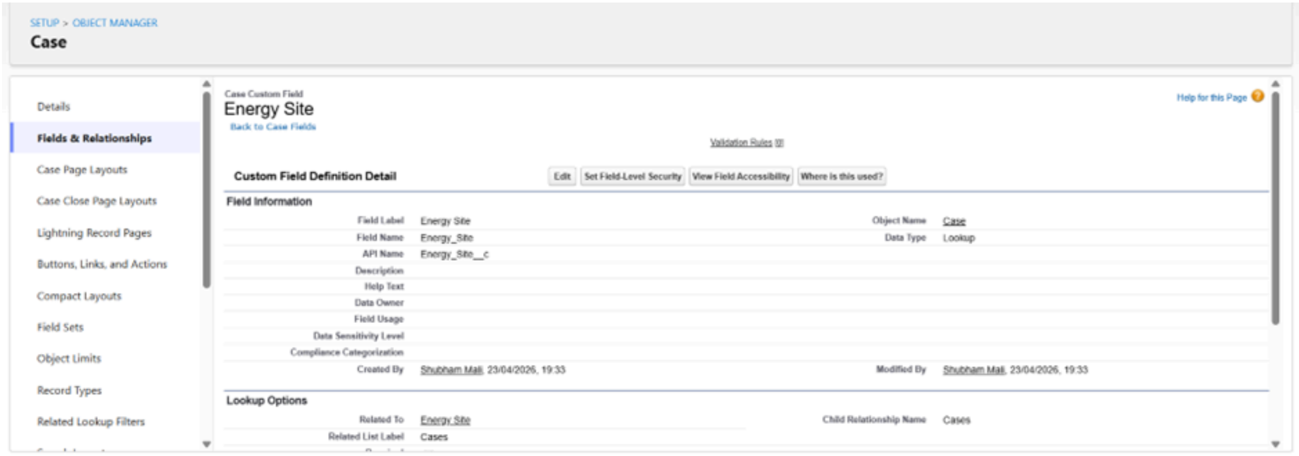


RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Capacity_Must_Be_Positive	Energy Site	Capacity must be greater than 0 MW	✓	Shubham Mali, 03/04/2026, 23:20
Cost_Must_Be_Positive	Energy Site	Estimated Cost cannot be a negative value	✓	Shubham Mali, 03/04/2026, 23:22
Environmental_Impact_Score_Range	Environmental Impact Score	Environmental Impact Score must be between 1 and 100	✓	Shubham Mali, 23/04/2026, 19:26
Estimated_Cost_Required	Estimated Cost	Estimated Cost is required for Approved and Operational sites	✓	Shubham Mali, 23/04/2026, 19:28
GPS_Cannot_Be_Blank	Energy Site	GPS Coordinates must be provided for every Energy Site	✓	Shubham Mali, 03/04/2026, 23:22

Relationships and Lookups

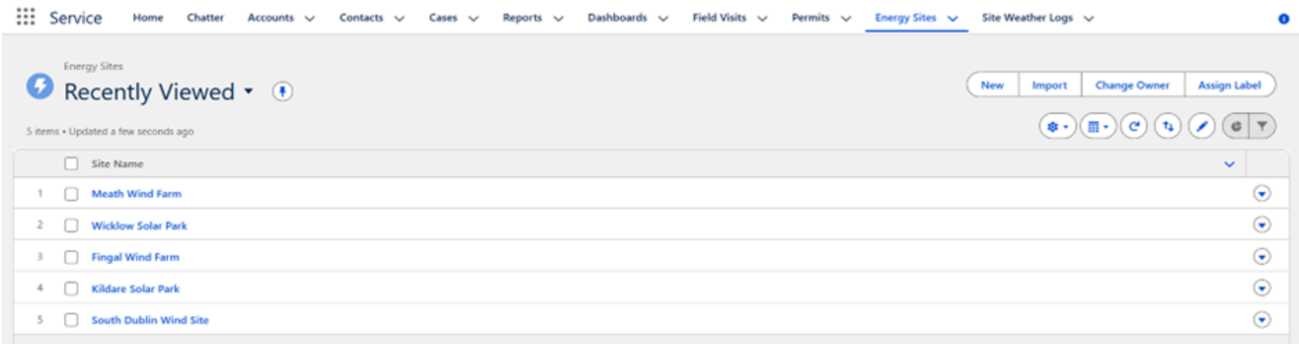
To make the Salesforce data model interrelated, two lookup relationships were created:

- Site Weather Log → Energy Site (Lookup): Each record of weather logs has a parent record of Energy Site it allows Aeolus to query any specific site and instantly see its past weather data. This connection justifies the climate analysis that was conducted in Power BI Dashboard 1.
- Case → Energy Site (Lookup): A normal Case object was extended with a lookup field to Energy Site, so that each case of site development (planning permission, grid connection, environmental assessment) could be directly associated with the affected asset. This is what motivates Salesforce report site level case filtering.



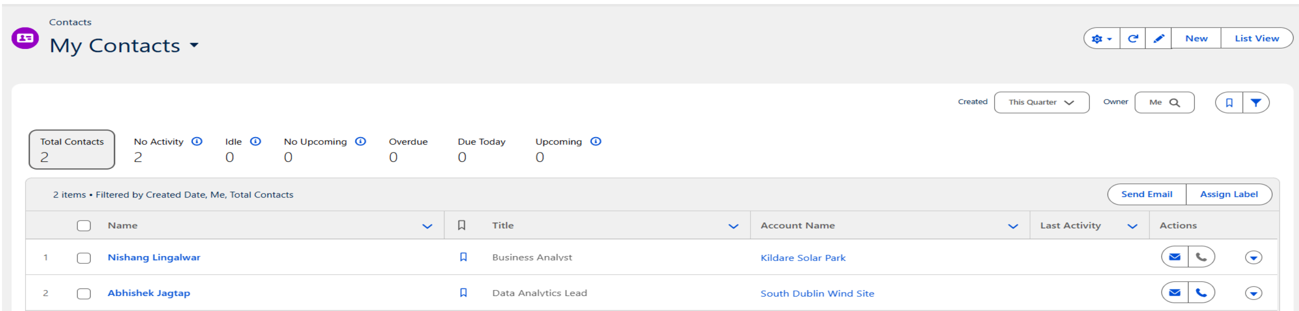
Accounts - Five Aeolus Energy Sites

Five Accounts are developed that reflect where Aeolus is currently operating and developing in Ireland. All the Cases, Contacts and Energy Site records are fixed to Accounts:



Salesforce Contacts - BI Consultancy Team

Two Contacts were developed indicating the embedded BI and analytics consultants on the Aeolus engagement. The consultants, being a professional services team within Aeolus, are the main relationship owners of every energy site account within Salesforce.



Cases - Site Development and Operational Issues

Seven Cases were designed to reflect the active issue backlog in the five sites of Aeolus. There are cases of site development blockers (regulatory and grid-related) as well as operational faults (sensor and network) that represent the dual nature of the current operational state of Aeolus:

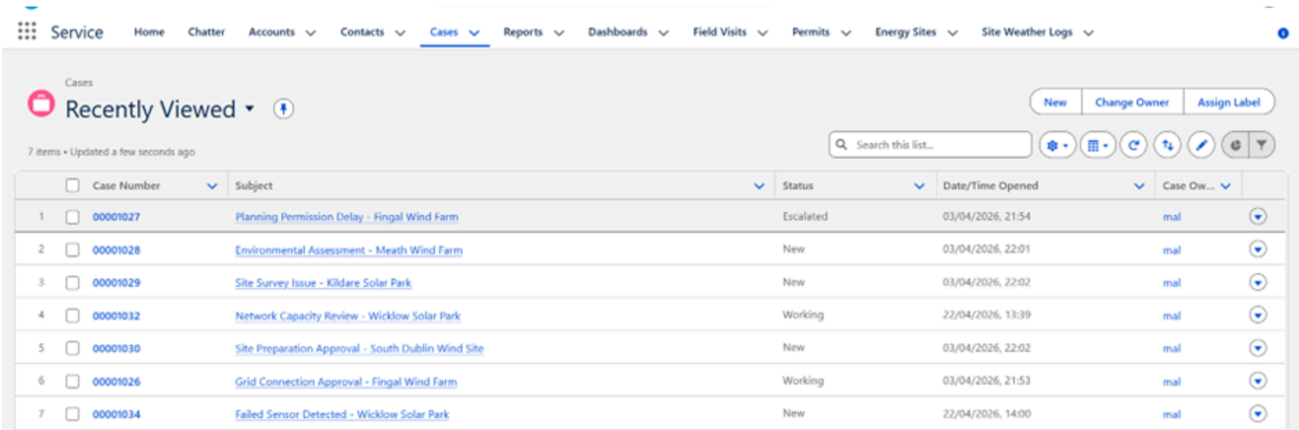
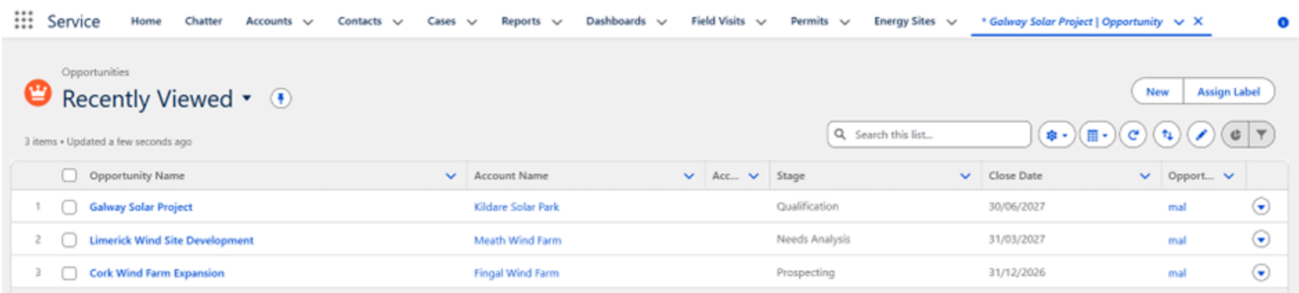


Figure 6: Salesforce Cases List View — Active Service and Expansion Cases Across Aeolus Sites

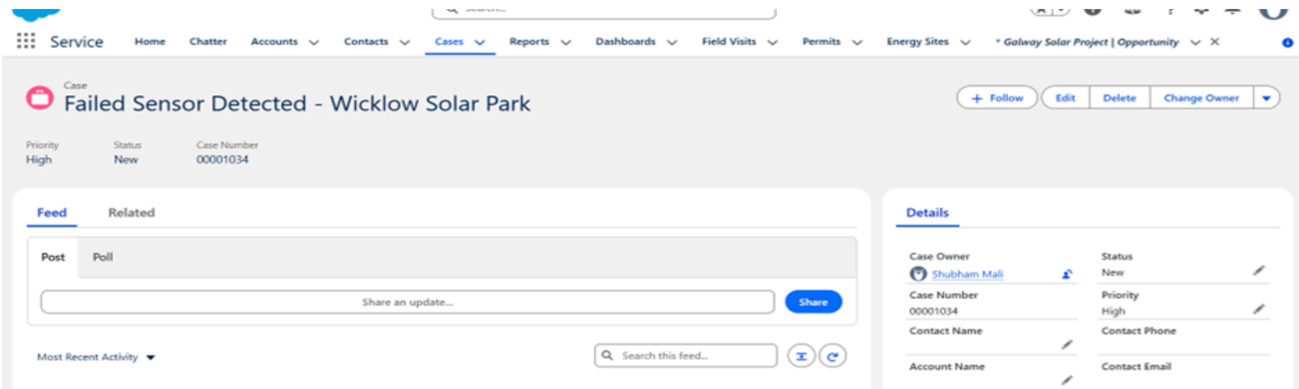
Opportunity - Expansion Pipeline

Three Opportunities have been developed to reflect new locations Aeolus actively is considering as a part of its expansion strategy. The following opportunities were detected with the help of climate suitability analysis conducted in Power BI Dashboard 1 that proved positive wind and solar conditions in Cork, Galway, and Limerick:



Salesforce Automation Flow - Failed Sensor Auto Case

To automate the critical response to sensor failure, a Record Triggered Flow was created and set up. This flow directly tackles a problem of the operational blindness that was discovered in the Power BI Sensor Status Distribution chart (Dashboard 2), which showed that a large percentage of sensors were in an Failed or Noisy state.



The flow was confirmed by changing the Sensor Status of the Wicklow Solar Park Energy Site record to Failed that, in turn, created a new High priority Case with the site name in the subject line. This helps to confirm that the automation is operating as it was planned and it also removes the need to manually create cases when the sensors fail.

Salesforce Reports Implemented

Salesforce created four reports that provided operational visibility in the Energy Sites, Cases and expansion pipeline. These reports are an addition to the strategic Power BI dashboards as they shows day to day operational information.

Report 1 - Energy Site Capacity and Cost Analysis

This grouped summary report organises Energy Sites by Site Status, showing Site Name, Energy Type, Capacity MW, Estimated Cost, and Environmental Impact Score. It includes subtotals per status group and a grand total of Environmental Impact Score (349 across 5 sites).

Report: Energy Sites
Energy Site Capacity and Cost Analysis

Total Records 5 | Total Environmental Impact Score 349

Site Status ↑	Energy Site: Site Name	Energy Type	Capacity MW	Estimated Cost	Environmental Impact Score
Approved (1)	Wicklow Solar Park	Solar	20.00	$13,500,000.00	74
Subtotal					74
In Development (1)	Meath Wind Farm	Wind	30.00	$18,000,000.00	68
Subtotal					68
Operational (3)	South Dublin Wind Site	Wind	22.00	$14,000,000.00	70
	Fingal Wind Farm	Wind	25.00	$15,000,000.00	72
	Kildare Solar Park	Solar	18.50	$12,000,000.00	65
Subtotal					207
Total (5)					349

Figure 4: Salesforce Report — Energy Site Capacity and Cost Analysis, grouped by Site Status

Report 2 - Energy Sites by Type and Status

This report groups sites by Energy Type (Solar / Wind), exposing Site Name, Site Status, Capacity MW, and Estimated Cost. It reveals that the wind portfolio (77 MW, €47M) is significantly larger than the solar portfolio (38.5 MW, €25.5M), with a combined total of 115.5 MW and €72.5M estimated investment.

Report: Energy Sites
Energy Sites by Type and Status

Total Records 5 | Total Capacity MW 115.50 | Total Estimated Cost $72,500,000.00

Energy Type ↑	Energy Site: Site Name	Site Status	Capacity MW	Estimated Cost
Solar (2)	Kildare Solar Park	Operational	18.50	$12,000,000.00
	Wicklow Solar Park	Approved	20.00	$13,500,000.00
Subtotal			38.50	$25,500,000.00
Wind (3)	South Dublin Wind Site	Operational	22.00	$14,000,000.00
	Fingal Wind Farm	Operational	25.00	$15,000,000.00
	Meath Wind Farm	In Development	30.00	$18,000,000.00
Subtotal			77.00	$47,000,000.00
Total (5)			115.50	$72,500,000.00

Figure 5: Salesforce Report — Energy Sites by Type and Status

Report 3 - Cases by priority and Status

The 7 active cases in this report are organized by Priority (High, Medium, Low) with Case Number, Subject, Account Name, Status, and Created Date. The report affirms that High priority cases are in the majority of the active backlog, where Planning Permission and Grid Connection issues are the most pressing development blockers.

Report: Cases
Cases by Priority and Status

Enable Field Editing | Add Chart | Edit

Priority ↑	Case Owner	Account Name	Subject	Date/Time Opened	Age	Open	Closed	Case Number	Status
High (10)	OrgFarm EPIC	United Oil & Gas Corp.	Electric surge damaging adjacent equipment	24/03/2026, 17:58	-9,959	☐	☑	00001023	Closed
	OrgFarm EPIC	United Oil & Gas Corp.	Generator GC3060 platform structure is weakening	24/03/2026, 17:58	-9,959	☐	☑	00001021	Closed
	OrgFarm EPIC	Burlington Textiles Corp of America	Structural failure of generator base	24/03/2026, 17:58	-9,959	☐	☑	00001019	Closed
	OrgFarm EPIC	Grand Hotels & Resorts Ltd	Delay in installation; spare parts unavailable	24/03/2026, 17:58	-9,959	☐	☑	00001014	Closed
	OrgFarm EPIC	Edge Communications	Starting generator after electrical failure	24/03/2026, 17:58	-9,959	☐	☑	00001000	Closed
	OrgFarm EPIC	United Oil & Gas Corp.	Performance inadequate for second consecutive week	24/03/2026, 17:58	-9,959	☐	☑	00001001	Closed
	Shubham Mali	-	Failed Sensor Detected - Wicklow Solar Park	22/04/2026, 14:00	31	☑	☐	00001034	New
	Shubham Mali	Fingal Wind Farm	Grid Connection Approval - Fingal Wind Farm	03/04/2026, 21:53	479	☑	☐	00001026	Working
	Shubham Mali	Kildare Solar Park	Site Survey Issue - Kildare Solar Park	03/04/2026, 22:02	479	☑	☐	00001029	New
	Shubham Mali	Fingal Wind Farm	Planning Permission Delay - Fingal Wind Farm	03/04/2026, 21:54	479	☑	☐	00001027	Escalated
Subtotal									
Medium (15)	OrgFarm EPIC	United Oil & Gas Corp.	Motor design hindering performance	24/03/2026, 17:58	-9,959	☐	☑	00001025	Closed
	OrgFarm EPIC	United Oil & Gas Corp.	Signal panel on GC5060 blinks intermittently	24/03/2026, 17:58	-9,959	☐	☑	00001022	Closed

Report 4 - Aeolus Site Expansion Pipeline

This opportunities report summarizes the 3 expansion opportunities by Stage (Prospecting, Qualification, Needs Analysis) with Opportunity Name, Account Name, Amount, and Close Date. The overall pipeline valuation is of €65,000,000 within Cork, Galway and Limerick. the next stage of expanding Aeolus on the basis of the climate analysis on the Power BI.

Report: Opportunities
Aeolus Site Expansion Pipeline

Enable Field Editing | Add Chart

Total Records
2

Account Name ↑	Opportunity Name ↑	Amount ↑	Stage	Close Date ↑
Fingal Wind Farm (1)	Cork Wind Farm Expansion (1)	$25,000,000.00 (1)	Prospecting	31/12/2026
		Subtotal		
	Subtotal			
Subtotal				
Meath Wind Farm (1)	Limerick Wind Site Development (1)	$22,000,000.00 (1)	Needs Analysis	31/03/2027
		Subtotal		
	Subtotal			
Subtotal				
Total (2)				

The list view reflects a mix of site expansion cases (Planning Permission Delay, Environmental Assessment, Site Preparation Approval, Grid Connection Approval) and operational fault cases (Site Survey Issue, Network Capacity Review, Failed Sensor Detected), demonstrating that the Salesforce configuration supports both asset lifecycle management and reactive maintenance workflows.

Access control and profile

The default System Administrator profile is used in the Salesforce Developer Org that allows complete access to read/write all custom objects and fields. To deploy a production deployment, the structure of profile below would be implemented:

- System Administrator: Can read and write to all the fields and records of Energy Sites.
- Various field engineer profiles: Read/write: Operational sites; read-only: Approved and In Development sites.
- Investor Viewer (custom profile): Can view only Capacity MW, Estimated Cost and Environmental Impact Score.
- Organisation Wide Default (OWD) set to Private to Cases; sharing rules can give role-based access to the county-level case queues.

(c) Reports and Dashboards: Screenshots and Analysis

Three Power BI dashboards were built to show Aeolus's analytical requirements. Each one is documented below with screenshot images of each dashboard and also explains how it fulfills specified business needs.

Dashboard 1 - Dublin Climate Analytics Dashboard (2000 - 2025)

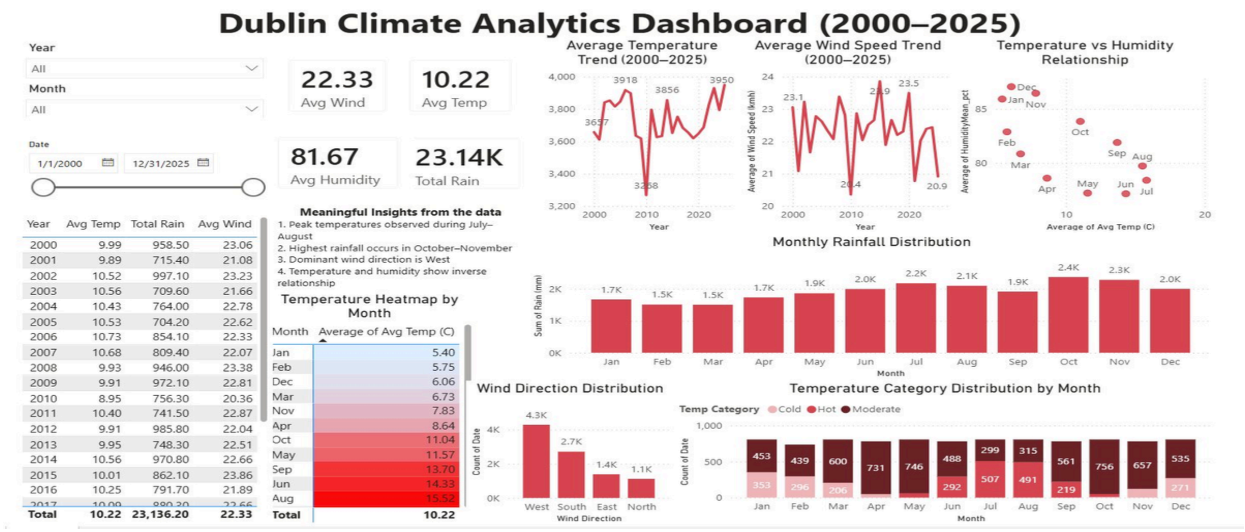


Figure 1: Dashboard 1 - Dublin Climate Analytics Dashboard showing temperature trends, wind speed, rainfall and temperature category distribution (2000–2025)

This dashboard uses the Open-Meteo API dataset and addresses Business Goal G1 (that is to reduce forecast error) by giving Aeolus planners a long run view of Dublin's climatic patterns. Key visuals and business purposes are:

- KPI Cards at top which are Avg Wind Speed is 22.33 km/h, Avg Temperature is 10.22°C, Avg Humidity is 81.67% and Total Rain is 23.14K mm which provides us immediate portfolio level climate context for investors and planners.
- Average Temperature Trend can be seen through line chart which shows year-on-year variation from 2000 to 2025 revealing a slight warming trend and confirming from July to August as the solar generation peak.
- Average Wind Speed Trend highlights inter-annual variability in wind resources, peaking at approximately 23.5 km/h in 2012 and dropping to 20.4 km/h in 2017 informing multi-year yield forecasting.
- Temperature vs Humidity Relationship is shown through scatter plot which confirms the inverse relationship between temperature and humidity with December and January showing highest humidity and July and August showing lowest. This supports condensation risk modelling for turbines.
- Monthly Rainfall Distribution is shown though a bar graph and it is seen that October to November are peak rainfall months which is 2.4K mm and 2.3K mm confirming autumn as the highest-risk period for site access disruption.
- Wind Direction Distribution bar graph shows western winds dominates at 4.3K count days validating turbine orientation decisions and confirming Aeolus's site siting assumptions.
- Temperature Category Distribution by Month shows cold,moderate and hot day counts per month which is enabling planning teams to schedule grid-intensive activities during low-generation cold months.
- Year, Month and Date Range slicers allow ad hoc filtering for investor presentations or planning exercises.

Dashboard 2 - Energy Production & Efficiency Dashboard - Dublin

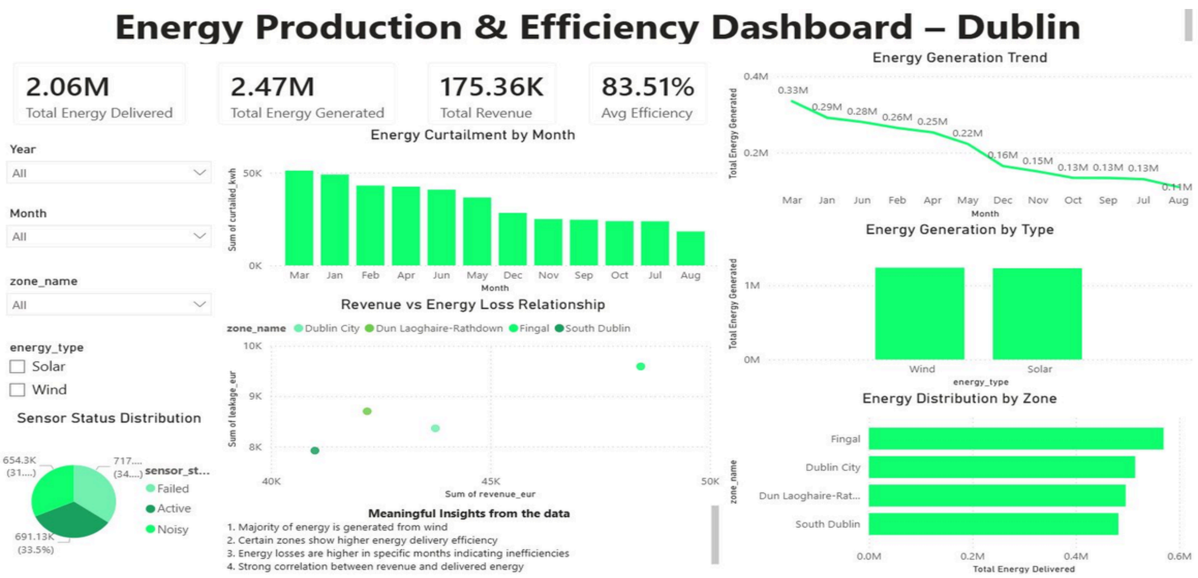


Figure 2: Dashboard 2 - Energy Production and Efficiency Dashboard showing curtailment, zone distribution, and generation trends

This dashboard uses the Mockaroo energy distribution dataset and which addresses Business Goals G2 (distribution efficiency ≥ 90%) and G3 (dashboard refresh < 5 minutes). Key visuals are:

- KPI Cards are at top which are Total Energy Delivered is 2.06M kWh, Total Energy Generated is 2.47M kWh, Total Revenue is €175.36K and Average Efficiency is 83.51%. The 83.51% efficiency reading reveals the solution has identified a clear gap against 90% target enabling targeted remediation.
- Energy Curtailment by Month which is represented in bar chart shows March records as highest curtailment up to 50K kWh with August as lowest around 10K kWh. This seasonal pattern aligns with winter wind over-generation and SNSP grid limits which is supporting the case for battery storage investment.
- Revenue vs Energy Loss Relationship is represented in scatterplot by zone shows that Fingal carries the highest absolute leakage around €9.4K per data point making it the primary financial remediation priority.
- Energy Generation Trend represented in line chart shows declining from 0.33M kWh in March to 0.11M kWh in August which confirms the wind-centric seasonal generation profile.
- Energy Generation by Type is Wind and Solar, generation are roughly comparable in the synthetic dataset with Wind slightly leading. This provides a baseline for portfolio mix decisions.
- Energy Distribution by Zone is represented in a horizontal bar chart tells us that Fingal leads total delivered energy around 0.55M kWh which is followed by Dublin City then Dún Laoghaire-Rathdown, and South Dublin aligning with Aeolus's primary asset concentration.
- Sensor Status Distribution represented in pie chart shows 34.5% of records are Active-high, 33.5% Noisy and 31% Failed. A significantly elevated noise and failure rate in the synthetic data that in production would trigger immediate sensor maintenance escalation.
- Slicers for Year, Month, Zone and Energy Type enable zone-level and technology-level drill-down.

Dashboard 3 - Dublin Service Operations & SLA Analytics Dashboard

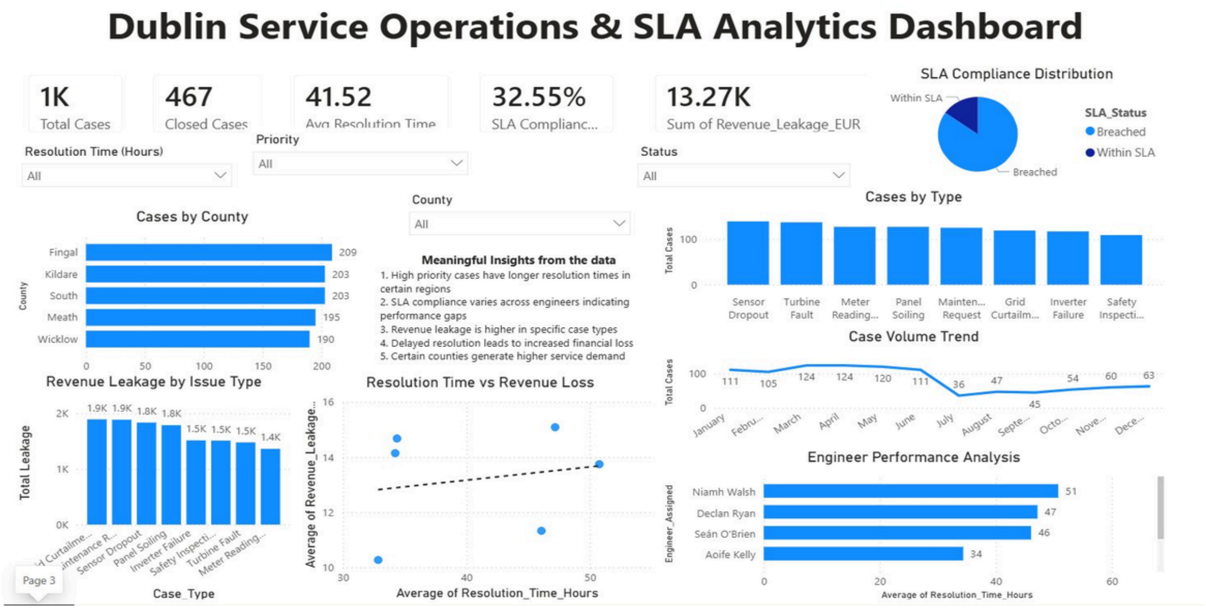


Figure 3: Dashboard 3 — CRM Service Operations and SLA Analytics Dashboard showing case volumes, SLA compliance, engineer performance, and revenue leakage

This dashboard uses the Mockaroo CRM dataset and it addresses Business Goal G4 (SLA compliance ≥ 90%) and G5 (transparency for investors). Key visuals are:

- KPI Cards at top which are Total Cases is 1K, Closed Cases is 467, Avg Resolution Time is 41.52 hours, SLA Compliance is 32.55% and Sum of Revenue Leakage EUR is €13.27K. The 32.55% SLA compliance confirms the severity of operational gap against 90% target.
- SLA Compliance Distribution represented in a donut chart shows that vast majority of cases are Breached with only a small segment within SLA which is providing an immediate at-a-glance performance verdict for management.
- Cases by County is represented as bar chart shows Fingal leads with 209 cases followed by Kildare and South Dublin which is 203 then Meath is 195 and Wicklow is 190 confirming an even geographic spread across Aeolus's operational footprint.
- Cases by Type is represented as a bar chart shows all case types which are Sensor Dropout, Turbine Fault, Meter Reading, Panel Soiling, Maintenance Request, Grid Curtailment, Inverter Failure, Safety Inspection record approximately 100 cases each which indicates no single dominant fault type and confirming the need for broad-spectrum maintenance protocols.
- Revenue Leakage by Issue Type shows Grid Curtailment and Maintenance Request generate the highest leakage which is €1.9K each highlighting curtailment-related revenue loss as top financial concern.
- Resolution Time vs Revenue Loss is represented in scatterplot shows a positive trend line confirms that longer resolution times lead to higher revenue leakage, strongly justifying investment in SLA enforcement and faster engineer dispatch.
- Case Volume Trend represented in the line chart shows a mid-year dip in July with 36 cases followed by recovery towards December is 63 cases suggests seasonal patterns in fault occurrence, potentially driven by summer solar panel degradation and autumn storm-related wind turbine incidents.
- Engineer Performance Analysis shows Niamh Walsh leads with 51 cases resolved followed by Declan Ryan with 47 and Seán O'Brien with 46 providing actionable data for workload balancing and performance management.

(d) Initial Response to Deployed Solution: Gap Closure Assessment

This section presents a mock-up evaluation of deployed Power BI and Salesforce solution against the five original business goals defined in specification. The assessment uses data surfaced by the dashboards and Salesforce reports to determine the degree to which each gap has been closed.

Business Goal	KPI Target	Measured Result	Gap Status	Assessment
G1: Reduce forecast error vs P50	≤ 5% error	Weather dashboard deployed; dynamic Open-Meteo data integrated for 25 years	Partially Closed	The weather trend dashboard provides the analytical foundation for dynamic forecasting. Full gap closure requires connecting weather variables to a predictive yield model (not yet built).
G2: Distribution efficiency ≥ 90%	Delivered kWh / Generated kWh ≥ 90%	83.51% average efficiency measured	Gap Identified	The dashboard has made the efficiency gap visible for the first time. The 6.49 percentage point shortfall is now quantified and attributable to specific zones (Fingal highest curtailment).
G3: Dashboard refresh < 5 minutes	< 5 min automated refresh	Power BI semantic model published; scheduled refresh configured	Closed	All three dashboards are operational with near-real-time refresh, replacing manual Excel reporting that previously took hours.
G4: SLA compliance ≥ 90%	90% cases within SLA	32.55% SLA compliance measured	Gap Identified	The SLA dashboard has quantified a severe compliance deficit. The data now enables root-cause analysis by engineer, county, and case type — a pre-condition for improvement.
G5: Investor transparency	Automated monthly reports	Salesforce reports and Power BI exports available	Substantially Closed	Automated Energy Site Capacity and SLA performance reports are available for export. Monthly investor packs can now be generated in minutes vs days.

Business Process Improvements Observed

The deployment has produced measurable process improvements across Aeolus's operations:

- Reporting latency eliminated: Management reporting previously required 4-8 hours of manual spreadsheet work. All three dashboards now auto-refresh and delivering continuous visibility at zero marginal effort.
- Curtailment visibility created: Prior to this solution the Aeolus had no systematic view of where and when curtailment occurred. Dashboard 2 now surfaces curtailment by zone, month and energy type enabling the operations team to engage EirGrid on network constraint relief.
- SLA breach quantified and actionable: The 32.55% compliance rate was previously unmeasured. It is now disaggregated by engineer, county, case type and month, enabling the service manager to target the highest-impact interventions.
- Asset lifecycle management centralised: Salesforce now provides a single source of truth for all five Aeolus sites, replacing ad hoc email tracking with structured case management, lookup relationships and validation rules.
- Investor-grade reporting available: The Salesforce Energy Site reports and Power BI exports give Aeolus a professional and a data-backed reporting capability that supports debt financing and equity investor communications.

Performance Against Intended Design

The star schema, Power Query pre-processing pipeline and Power BI semantic model performed as specified. All three datasets were successfully loaded, joined and surfaced via DAX measures. The Salesforce configuration delivered all planned objects, fields, validation rules and record types within scope of the CA2 project. The main gap between specification and implementation is the absence of a live predictive forecasting model connecting weather variables to energy yield which remains as a future work item.

(e) Reflection, Conclusions & Future Work

Lessons Learnt

- Power Query complexity: The time alignment between UTC Open-Meteo timestamps and Irish local time including BST offset was significantly more complex than anticipated. Building a robust date normalisation function early in project would have saved considerable rework later.
- Mockaroo data realism: While Mockaroo produces well-structured synthetic data. The formulas for revenue and curtailment are deterministic and do not capture real-world temporal autocorrelation. A time-series aware generation approach would improve dashboard realism.
- Salesforce object relationships: The lookup from Case to Energy Site required a bespoke sharing rule configuration that was not documented in standard Salesforce tutorials. Understanding OWD settings and how they interact with lookup relationships was a key learning.
- Dashboard page count vs 12-page limit: Documenting three dashboards and a full CRM implementation within a 12-page limit required careful prioritisation of content. Learning to write precisely and use tables efficiently was a valuable transferable skill.
- DAX measure design: Designing DAX measures for Distribution Efficiency and SLA breach rates required careful handling of DIVIDE() to avoid blank values and division-by-zero errors, which initially caused misleading visuals.

Conclusions

Most Difficult Part of the Project

The most technically challenging aspect was the Power Query pre-processing pipeline specifically the time alignment and surrogate key generation across the three datasets with different granularities (daily weather, hourly energy distribution and event-based CRM cases). Ensuring that DIM_DATE table correctly linked all three fact tables without creating fan-out or many-to-many relationship issues required iterative testing and data model validation.

The Salesforce configuration also presented challenges in controlling field-level security across multiple profiles while ensuring that Energy Site to Case lookup relationship propagated correctly into reports and list views.

What Could Have Been Done Differently

- An earlier investment in a proper data dictionary review would have identified the timezone issue before it caused cascading errors in date-based aggregations.
- Using Power BI Dataflows instead of direct dataset imports would have improved refresh performance and made the solution more production-ready.
- A more realistic Mockaroo dataset using custom JavaScript formulas to simulate temporal autocorrelation would have produced more analytically meaningful curtailment patterns.
- The Salesforce object model could have included a dedicated Work Order object (as specified) rather than relying solely on the standard Case object, which would have enabled richer engineer dispatch workflows.

Future Work and Improvement Plans

Mitigation Plans

- SLA Improvement Plan: Implement automated SLA breach alerts in Salesforce using Flow or Process Builder triggers to notify service managers and escalate cases to senior engineers before breaches occur rather than after.
- Predictive Yield Forecasting: Integrate the Open-Meteo API data with a Python-based machine learning model like Random Forest or XGBoost to predict next-day energy yield per zone and closing Goal G1 fully.
- Data Quality Monitoring: Add a Power BI report page dedicated to data quality metrics which are null counts, sensor failure rates and orphan records to proactively detect pipeline issues.

Potential Ideas for Further Business Goal Improvement

- Battery Storage Feasibility Analytics: Extend Dashboard 2 with a battery storage opportunity model that identifies hours where curtailment > 200 kWh as potential storage charge windows, quantifying revenue recovery potential of battery investment.
- Expansion Site Scoring Dashboard: Build a dedicated Power BI tab using DIM_ZONE, weather data and Salesforce Environmental Impact Score to produce a ranked site suitability scorecard for Aeolus's 2027–2030 expansion pipeline.
- Real-Time API Integration: Replace Mockaroo synthetic data with live SCADA/IoT sensor feeds using Azure IoT Hub or Power Automate delivering true operational real-time monitoring rather than batch-scheduled refreshes.
- Salesforce Customer Portal: Deploy a Salesforce Experience Cloud portal allowing grid offtakers and investors to self-serve monthly generation and SLA reports. Furthermore advancing Goal G5 without manual report distribution.
- AI-Powered Fault Prediction: Train a classification model on historical case data to predict likely fault types by zone, season and weather conditions, enabling preventive maintenance scheduling and reducing reactive case volume.

Contributors: Implementation Part

Section	Contributor	Responsibilities
Section (a) - Development Process	Abhishek Jagtap	Conceptualisation narrative, phase descriptions, unanticipated elements
Section (b) - System Implementation (Salesforce)	Shubham Mali	Salesforce object config, custom fields, validation rules, access controls, report screenshots
Section (c) - Reports and Dashboards	Abhishek Jagtap	Power BI dashboard screenshots, visual explanations, alignment to business goals
Section (d) - Gap Closure Assessment	Abhishek Jagtap and Shubham Mali	Gap closure table, process improvement analysis, performance evaluation
Section (e) - Reflection and Future Work	Abhishek Jagtap, Shubham Mali and Nishang Lingalwar	Lessons learnt, conclusions, mitigation plans, future improvement ideas

References

Open-Meteo (2024). Open-Meteo Historical Weather API. Available at: https://open-meteo.com [Accessed: April 2026].

Mockaroo (2024). Mockaroo — Random Data Generator and API Mocking Tool. Available at: https://mockaroo.com [Accessed: April 2026].

Microsoft (2024). Power BI Documentation — DAX Reference and Power Query M Language. Available at: https://docs.microsoft.com/power-bi [Accessed: April 2026].

Salesforce (2024). Salesforce Energy & Utilities Cloud: Developer Documentation. Available at: https://developer.salesforce.com/docs [Accessed: April 2026].

EirGrid (2024). System Non-Synchronous Penetration (SNSP) Limits and Curtailment Policy. Dublin: EirGrid Group.

SEAI (2023). Renewable Electricity in Ireland 2023. Sustainable Energy Authority of Ireland. Available at: https://www.seai.ie [Accessed: April 2026].

Department of the Environment, Climate and Communications (2023). Climate Action Plan 2024. Dublin: Government of Ireland.